PROJECT EXPLAINER | HINGLISH

Decompile 3D

PCB photo se interactive 3D learning experience tak

Ek beginner-friendly web prototype jo PCB ki photo ko AI se approximate component map mein badalta hai, us map ko interactive 3D board par dikhata hai, aur electronics seekhne ke liye tutor aur sample waveform deta hai.

PROJECT TYPE AI-assisted educational web app	**FRONTEND** React + Three.js
BACKEND Python + FastAPI	**AI SERVICES** Gemini, Groq, OpenRouter

Is guide mein purpose, technologies, architecture, code map, AI flow, run karne ke steps, limitations aur future roadmap diya hai.

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1. Project kya hai?

Simple explanation: Decompile 3D ek web-based PCB learning and visualization prototype hai. User board ki photo upload karta hai; AI visible components aur unki approximate position ka structured JSON banata hai; browser us data se interactive 3D board, photo overlay, sample waveforms aur chat-based electronics tutor dikhata hai.

Kaunsi problem solve karne ki koshish hai?

PCB ko pehli baar dekhne wale student ya hobbyist ko chhote components, unke labels aur possible roles samajhna mushkil hota hai. Flat photo mein depth, component focus aur explanation saath nahi milte. Project in cheezon ko ek hi interface mein laane ki koshish karta hai.

Solution ko chaar parts mein samjho

- **Photo recognition:** Gemini photo dekh kar component type, label, size aur board ke andar approximate x/y position return karta hai.
- **3D visualization:** React Three Fiber board aur components ko Three.js scene mein render karta hai; user rotate, zoom aur part select kar sakta hai.
- **Learning aids:** selected part ka short role, us type ki sample waveform aur AI tutor ka jawab dikhaya jata hai.
- **Hologram/flow:** light, dark aur holographic themes, solder details aur illustrative animated flow paths.

Project category aur intended users

Category	Description
Application	Interactive browser-based educational demo / prototype
Domain	Electronics learning, PCB visualization, AI-assisted explanation
Primary users	Students, makers, hobbyists, repair learners, demo presenters
Input / output	Input: PCB image and chat questions. Output: estimated component map, 3D scene, explanations and illustrative signals.

Presentation mein ek line: "Ye app PCB photo ko interactive learning view mein badalti hai: AI component identify karne ki koshish karta hai aur 3D UI se user board ko explore karta hai."

2. Architecture aur end-to-end workflow

Frontend aur backend local development mein alag servers hain. Browser React app load karta hai. Vite `/api` requests FastAPI ko proxy karta hai. Backend AI provider se response lekar frontend ko bhejta hai.

Photo analysis data flow

1. User	2. React	3. FastAPI	4. Gemini	5. UI
PCB photo choose	Resize + base64	/api/analyze	Vision -> JSON	3D + markers

Tutor chat provider order

First	Agar unavailable/fails	Last fallback
Groq chat API	Gemini chat (key ho to)	OpenRouter (key ho to)

Gemini vision photo analysis ke liye use hota hai. Chat pehle Groq try karti hai; phir Gemini; aakhir mein optional OpenRouter. Provider models source code mein defaults ke roop mein hain, aur `.env` overrides support hote hain.

Shared data contract: board JSON

AI response mein `board` (name, aspect) aur `parts` list hoti hai. Har part mein type, label, value, x, y, w, h aur role hota hai. Frontend isi `spec` ko 3D model, photo markers, component chips aur tutor context mein reuse karta hai.

Position meaning: x/y/w/h board bounding box ke percentages hain (0-100). x/y part ka center hai; w/h approximate footprint. Backend type allow-list check karta, numbers clamp karta, unknown types ko `other` banata aur invalid aspect ratio ko default karta.

Important boundary: app photo se copper tracks, schematic ya verified electrical netlist recover nahi karti. Board positions aur flows estimates hain.

3. Photo se 3D tak: implementation

Step	Code mein kya hota hai	User ko kya dikhta hai
1. Sample	Startup par `/api/sample` initial demo spec deta hai.	Backend connect ho to demo board.
2. Photo pick	Browser canvas longest side max 1024 px karta; JPEG quality 0.85 par data URL banata hai.	Upload chhota aur API ke liye ready.
3. Analyze	POST `/api/analyze` with `{image: ...}`; Vite request ko port 8000 par forward karta hai.	Analysis status message.
4. Gemini	FastAPI image + prompt Gemini `generateContent` endpoint ko bhejta; JSON format maangta hai.	AI visible board ka estimate.
5. Clean	Backend JSON parse karta, part values normalize karta, empty result par error deta.	Frontend ko normalized `spec`.
6. Render	React state update se Board3D, PhotoPane, chips aur Wave redraw hote hain.	3D board, photo boxes, waveform update.

3D geometry ka basic math

Board width fixed world scale se aati hai; board height `aspect` ratio se calculate hoti hai. Percent x/y ko centered world x/z coordinates mein map kiya jata hai. Part w/h se box ya cylinder footprint banta hai. Isliye model visual preview hai, manufacturing-ready CAD model nahi.

Photo overlay aur selection

Real photo available ho to wahi dikhayi jaati hai aur guessed percentages clickable boxes banate hain. Photo nahi ho to 2D top-view fallback hai. 3D model, photo markers aur component chips mein selected part sync hota hai; selection camera ko part par focus karti hai.

Chat ke saath board context

Chat request mein latest 10 messages, current board spec aur selected component bheja jaata hai. System prompt tutor ko student ki language match karne aur board-photo guesses ko uncertain batane ko kehta hai.

4. Frontend, 3D aur learning features

Technology	Is project mein role
React 18	Components, UI state, upload/chat handlers, conditional mobile tabs.
Vite 5	Frontend dev server, React plugin, `/api` proxy.
Three.js 0.160	3D geometry, camera, materials, scene rendering.
React Three Fiber 8	Three.js scene ko React components/JSX se describe karta hai.
Drei 9	OrbitControls, labels (Html), view gizmo, line and edge helpers.
CSS + Canvas 2D	Responsive interface and custom animated waveform plotting.

Board view mein kya-kya hai?

- Top, Front, Side, Iso camera presets; drag rotation, pan, scroll zoom; auto-rotate toggle; axis and labels.
- Light, Dark and Hologram display modes. Hologram uses custom GLSL shader: edge/Fresnel glow, scanlines, subtle flicker/glitch, grid, edge outlines and moving scan beam.
- Generic part geometry, labels/ref/value, component selection, approximate solder pads/leads and fullscreen browser control.
- Flow toggle: Power, Ground return, Selected component ya All. Color-coded PCB-style trace lanes ke saath moving pulses. Flows estimated component locations se bante hain; real board connectivity prove nahi karte.

Waveform feature

`Wave.jsx` Canvas 2D context aur `requestAnimationFrame` use karke sine, square, ramp, pulse aur related shapes draw karta hai. Har component type ke liye explanatory signal definitions hain. Ye live voltage/current reading, sensor capture ya circuit solver ka output nahi hai; teaching visual hai.

Reliable pitch: "App component ko samjhane ke liye generic signal animation deti hai; real board par measurement nahi karti."

5. Backend APIs aur code map

Backend routes

Route	Function	Input / output
GET `/api/status`	Configured API key providers ki availability.	No body -> true/false flags.
GET `/api/sample`	Default sample board.	No body -> board JSON.
POST `/api/analyze`	Gemini vision se image-to-spec.	Image string -> normalized spec/error.
POST `/api/chat`	Board-aware electronics tutor chat.	Messages/spec/selection -> reply/provider.

Important files

File	Responsibility
`backend/main.py`	FastAPI routes, env reading, provider calls/fallback, Gemini prompt, spec cleanup, chat prompt.
`backend/sample.json`	USB-C Li-ion charger/boost module naam wala demo spec aur sample components. Sample ko verified board data na bolen.
`backend/requirements.txt`	Python dependencies: FastAPI, Uvicorn standard extras, HTTPX.
`backend/.env.example`	API-key names and optional model override examples.
`frontend/src/App.jsx`	Main UI/state, sample load, image resize/upload, chat, selection, themes, flow selector, fullscreen.
`frontend/src/Board3D.jsx`	3D geometry, joints, labels, camera, hologram shader, scan/grid and flow animations.
`frontend/src/Wave.jsx`	Illustrative signal definitions and canvas animation.
`frontend/src/styles.css`	Layout, colors, mobile breakpoints, photo boxes, chat and flow panel.
`frontend/src/main.jsx`	React entry point; mounts App and global CSS.
`frontend/vite.config.js`	React plugin and backend `/api` proxy.
`frontend/package.json`	NPM scripts and React/Three ecosystem dependency versions.
`run.bat`, `.gitignore`	Windows launcher and ignored generated/secrets folders.

6. Run/setup aur configuration

Quick start - Windows

Project root mein `run.bat` double-click karna intended one-click flow hai. Backend port 8000, frontend port 5173, aur browser `http://localhost:5173` par khulta hai.

Manual setup

Backend terminal	Frontend - doosra terminal

`cd backend`
`..\.venv\Scripts\activate`
`pip install -r requirements.txt`
`copy .env.example .env`
`uvicorn main:app --reload --port 8000`
	`cd frontend`
	`npm install`
	`npm run dev`
	
	Browser: `http://localhost:5173`

Keys kis kaam aati hain?

- `GEMINI_API_KEY`: photo analysis; Gemini chat fallback ke liye bhi.
- `GROQ_API_KEY`: tutor chat ka first provider.
- `OPENROUTER_API_KEY`: optional last chat fallback.
- Optional model settings: `GEMINI_MODEL`, `GROQ_MODEL`, `OPENROUTER_MODEL`. API keys `.env` mein rakho; `.gitignore` env files ko exclude karta hai.

Launcher aur network note

Current `run.bat` startup se pehle port 8000/5173 ke matching process IDs ko force-kill karne ki koshish karta hai. `findstr :8000` substring match exact port tak limited nahi, aur kisi aur project ka process band ho sakta hai. Frontend Vite proxy `/api` ko localhost:8000 par bhejta hai.

High-level tools list

Python 3 + FastAPI + Pydantic + HTTPX; React 18 + Vite 5; Three.js + React Three Fiber + Drei; CSS; browser Canvas 2D; Gemini, Groq aur OpenRouter HTTP APIs. Python dependencies requirements mein hain; frontend dependencies `package.json` mein.

7. Limitations, future work, aur demo pitch

Abhi kya claim nahi karna chahiye

- Component labels, types, values, roles aur coordinates AI estimates hain; confidence scoring ya manual correction UI nahi.
- Gerber/KiCad/schematic/netlist parsing nahi; copper track connectivity infer/verify nahi hoti.
- 3D components generic shapes hain; exact package, pinout, board stackup, copper layers or manufacturing CAD nahi.
- Flow paths illustrative hain; signal simulation ya real current path nahi.
- Waveform actual test equipment, circuit telemetry ya SPICE engine se nahi aati.
- CORS `*` configured hai; auth, rate limits, strict upload validation, persistence aur production deployment hardening nahi dikhte. Public internet par daalne se pehle security work chahiye.

Future mein kaam kahan aa sakta hai?

Roadmap	Possible improvement	Benefit
Photo accuracy	Perspective correction, OCR, confidence, user edits.	Recognized parts ko user confirm/correct kare.
Real connectivity	KiCad/Gerber/netlist/schematic import.	Actual tracks/pins ke flows.
Engineering simulation	SPICE engine and user component values.	Waveforms physics/model se linked hon.
Richer board view	Datasheet 3D library, copper layers, underside and export.	Inspection and documentation useful ho.
Product readiness	Auth, validation, allowed CORS origins, limits, privacy and logging.	Safer public/organization deployment.

45-second explanation jo tum bol sakte ho

"Decompile 3D ek AI-assisted electronics learning prototype hai. Main PCB photo upload karta hoon; browser use resize karta hai aur FastAPI backend Gemini vision ko bhejta hai. Gemini component map ka JSON deta hai, jise React Three Fiber aur Three.js interactive board aur photo markers mein render karte hain. Part select karke uska role, us type ki sample waveform aur AI tutor se explanation dekh sakta hoon. Hologram aur flow visuals learning aids hain; ye actual netlist ya electrical measurement ka replacement nahi hain."

Demo ka order

1) Sample board load. 2) 3D rotate/zoom and select a part. 3) Hologram and Flows show. 4) Photo analysis only if Gemini key/network ready. 5) Tutor only if provider key configured. Waveform ko illustrative bolna.

Short summary: app ka strongest current role educational visualization aur demo hai. Verified PCB reverse engineering, circuit simulation aur hardware diagnosis future extensions hain.